# Smart Lab Portal — Status Report

BSc thesis (GIU) · 14 October 2026, night (update 3, with live camera measurements) · branch `offline-router-door-photos`, HEAD `802638b` + uncommitted work from two missions (performance; ESP32-CAM face speed)

> **Labels used: LIVE-VERIFIED** = measured on the real hardware/live system · **TESTED OFFLINE** = automated tests or simulation/recordings, no hardware · **UNVERIFIED** = not yet measured. No hardware result in this report is estimated; numbers come from logs and measurements.

## 1. Summary

- **UR5e digital twin** is LIVE-VERIFIED (read-only RTDE, 48 Hz, joint mapping checked joint by joint, stale detection 2.0 s).
- **FANUC ER-4iA digital twin** is LIVE-VERIFIED (read-only HTTP GET of `/MD/CURPOS.DG`, ~104 ms per read, ~3 Hz; the twin followed the pendant). The J2/J3 coupling still needs a pose test.
- **Camera 1** ran live at 7 analysis passes/s on the GPU. The 14 Oct tracking/identity overhaul is deployed but not yet confirmed live (the lab closed).
- **Camera 2 (MC200 1080p)**: software ready and tested with two simulated cameras; the device arrives tomorrow.
- **Performance**: a full simulated load test now exists. One real bottleneck was found and fixed (Command Center reloaded all its data on every live message); no backend or database bottleneck was found.
- **Face enrollment** (new): automatic capture from the door camera's normal stream, no camera round trip per sample; first sample 0.45 s after start in the offline test. A face that is clearly another enrolled person is refused. Not live yet: the face server needs a restart (your approval).
- **Door recognition**: the laptop needs ~40–49 ms per frame; the camera's capture + Wi-Fi upload (~450–620 ms, earlier measurement) is the real delay. Threshold and door logic unchanged; 0 wrong-person matches on the stored samples.
- **ESP32-CAM firmware**: building with the installed Arduino-ESP32 core 3.3.12 breaks the camera. The board was restored from a full backup and works as before. The firmware fix stays in the source until an earlier core is installed.
- **Network**: the laptop is back on the Smart_Lab router by cable (192.168.1.200); everything runs locally, no internet. Live measurement: about **690 of every 780 ms** per camera frame is spent on the camera's own Wi-Fi link; the laptop needs ~90 ms.
- **Safety**: robots stay read-only, the door is opened only by the Master ESP32, production data untouched, nothing pushed.

## 2. Work completed (12–14 October)

### Robots

- UR5e collector (`tools/robot_monitor`, Python 3.12 + ur_rtde, receive only): connection check, validity check (6 finite joints), controller-timestamp de-duplication, movement events, home calibration, stale detection.
- FANUC source: only an HTTP GET of the current-position page; parser checked against the real controller page (R-30iB Mate Plus, HandlingTool V9.30); world-frame TCP only; motion detected on the

robot's own joints; the twin-only J3+J2 coupling applied only for drawing. A test proves the source class has no write path (no POST/PUT, sockets, KAREL, register writes).
- FANUC twin shows the Schunk EGP 40 electric gripper (drawn approximately). `start_fanuc_monitor.bat` and `launch.bat` start the monitor when `FANUC_IP` is set in the git-ignored `robot.env`.

**Security camera and identity**

- GPU (DirectML, GTX 1650): YOLO26s detector, one shared detector session and a single GPU lock — this fixed the `nvwgf2umx.dll` driver crash caused by a second DirectML session.
- Analysis loop: error handling, live fps, Windows QuickEdit disabled (clicking the console no longer freezes the camera), per-port guard against double processes.
- Tracking: thin duplicate boxes dropped; still people keep their track 6 s; appearance check after a gap. On real footage, broken tracks fell from 12 to 4.
- Identity rules: door entry, face, look, place, fresh-only elimination, home desk (shown only as a guess "? (desk)", never assigned on desk alone, yields to a newer door entry). Identity decisions logged per camera.
- Two cameras: per-camera env file, port, window title and logs; all `camera*.env` git-ignored; credentials masked in all output.

**Backend, portal and data**

- Environment "latest values" query rewritten (newest per lab and metric): query 72 → 2.1 ms, function 92 → 11.6 ms, live `/monitor` 97 → 59 ms (deployed).
- Test isolation: tests are forced onto `smartlab_test` with an advisory lock so two runs cannot deadlock.
- n8n workflows 01/03/05/07 recovered byte-exact and the repo copy aligned with the live one (live workflows not edited).
- Backups: snapshot branch, uncommitted-work patch, database dump before the deploy, n8n recovery folder, lab evidence screenshots.

**This session (performance mission)**

- New repeatable load test: `tools/perf/load_simlab.py`, `tools/perf/ws_clients.mjs`, guide `docs/perf/LOAD_TEST.md`. Runs on a separate backend copy and database (`smartlab_loadtest_test`, port 8100) — refuses the live portal.
- Fix in `frontend/src/pages/CommandCenter.tsx` (uncommitted, not deployed): see section 4.

## 3. Measurements (all SIMULATED load, isolated stack)

Workload for 120 s: 2 simulated robots (UR5e + FANUC models, ~9.5 telemetry posts/s), 2 camera services on recorded lab video (GPU), an environment node every 2 s, 3 WebSocket clients, and dashboard polling **faster than the real portal** (worst case). Live camera 1 kept running alongside.

| Endpoint (server side) | p50 ms | p95 ms | Note |
|---|---|---|---|
| /health | 13–26 | 30–92 | |
| /robots, /labs/hazards/active | 19–21 | 46–59 | |
| /robots/:id/events | 35 | 86 | |
| /security/cameras, /system/status | 29–43 | 71–99 | |

| | | | |
|---|---|---|---|
| /summary | 52 | 143 | |
| /monitor | 72–91 | 204–251 | with no browser open; 0.8–1.0 s p95 while a browser also loaded the stack |

- API errors: **0** in every run. WebSocket: 9.7 messages/s and 2.2 KB/s per client (robot_state 7.6/s, capped per robot).
- Two simulated cameras: 6.5–6.7 analysis passes/s each, pass p95 170–282 ms, frame age p95 84–95 ms, GPU ~65%, VRAM ~1.0 GB, system CPU ~69%.
- `/monitor` breakdown: ~22 ms idle (15 queries; the slowest query is 5.9 ms). Under load, measured from a separate process, it took ~51 ms: the extra time is the laptop's CPU being busy (cameras), not slow queries. The backend container used 20–56% of one core — not saturated.

## 4. Bottleneck found and fixed

> **Command Center (/command) reloaded all 8 of its data endpoints on every live status message** (camera state, environment update, robot status), after only a 400 ms delay; every robot status also re-fetched robot events immediately. With a camera, robots and the environment node reporting, the page reloaded about once per second.

- **Fix:** safety messages (hazard, alert, door) still refresh within 400 ms exactly as before. Status messages now refresh at most every 3 s, and the latest one is always applied (never dropped). Robot events follow the same 3 s limit. No alarm is throttled or discarded; the separate hazard alarm component is unchanged.
- **Result (TESTED OFFLINE, same load):** /command browser requests 13.5 → 9.4 /s; /monitor 0.8 → 0.4 /s; robot events 1.5 → 0.7 /s. Type check and production build pass.
- **Not yet deployed:** it goes live only after your approval of a commit and a `docker compose build frontend` + restart of the frontend container.

### Remaining performance notes

- /command shows long main-thread tasks in headless Chrome, but a CPU profile shows 12.4 of 20 s is native rendering: headless Chrome has no GPU, so the 3D twin runs in software WebGL. The page's own JavaScript is small (~0.6 s per 20 s). The twin's cost in a normal browser with a GPU is UNVERIFIED.
- The `/api/health` times measured inside the /command page (300–560 ms) are caused by that blocked main thread; the server answers the same call at p95 < 100 ms.
- Each robot telemetry post writes the robot's row and commits (~10/s for two robots). It is within capacity today; batching it would change how "last seen" works, so it was not changed.

## 5. ESP32-CAM face enrollment and door recognition

### Where the time goes (TESTED OFFLINE, the camera's own 640×480 frames)

| Stage on the laptop (per door frame) | median ms | p95 ms |
|---|---|---|
| JPEG decode | 2.5 | 5.0 |
| QR decode (fast) | 10.6 | 26.9 |
| Face detection (YuNet, 320 px) | 7.9 | 12.0 |
| Face alignment | 0.2 | 0.3 |

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|---|---|---|
| Face embedding (SFace) | 19.3 | 24.4 |
| **Whole /analyze request, real server code** | **48.9** | **61.8** |

- Earlier measurements at the door put the camera's capture + Wi-Fi upload + reply at about **450–620 ms per frame** (documented in the sketch). The laptop is ~10% of the time; the camera and the network are the bottleneck.
- **ONNX Runtime** was evaluated: identical embeddings (cosine 1.000000 on 40 samples), SFace 27 → 21 ms. About 6 ms of a ~500 ms frame, so not adopted (one more runtime to maintain for no visible gain).
- **YOLO26** was not used for faces: the face detector already takes ~8 ms, so it cannot measurably improve this stage. The face-recognition model (SFace) was not replaced.
- **Accuracy** (leave-one-out over the 74 stored samples of 5 people, threshold 0.363 unchanged): 72 right (97.3%), 2 refused, **0 wrong person**. Genuine cosine median 0.848; impostor median 0.189, impostor maximum 0.448 — above the threshold for one pair, but the right person still scored higher. The door also requires the face to match the booking/RFID identity.

### Live measurements, 14 Oct night (current firmware + current face server, all local)

| Step | Time | Label |
|---|---|---|
| Laptop → Smart_Lab router (cable) | 0 ms | LIVE |
| Face server work per camera frame (same frame, localhost, no face) | 89 ms avg, 134 max | LIVE |
| Camera's full frame trip, its own [PERF] log | **782–795 ms** → 1.2–1.4 frames/s, 8 KB/frame, Wi-Fi −21 dBm | LIVE |
| Ping laptop → camera (over the cable, then router Wi-Fi) | 138–324 ms (avg 217) | LIVE |
| Camera `/status` over HTTP | avg 680 ms (132–1565) | LIVE |
| **Enrollment preview, old way** (camera `/snapshot`) | **avg 1410 ms**, max 3.5 s | LIVE |
| **Enrollment preview, new way** (frame already on the face server) | **~2 ms** | LIVE |

- Routing checked: the laptop reaches the camera over Ethernet (metric 10), not the Vodafone Wi-Fi. The wired hop is 0 ms, so the delay is between the router and the camera.
- The camera sits right next to the router at an unusually strong −21 dBm. Signals that strong can overload the ESP32 receiver and cause retransmissions; Wi-Fi power saving would also look like this, although the sketch has switched it off since the first commit. **Next test:** move the camera 2–3 m away and re-measure — no code change needed if that is the cause.

### What changed (uncommitted)

- **Automatic enrollment** (face server `/enroll/auto`, device key required). Samples come from the frames the door camera already sends (~6/s). A frame is saved only if it has exactly one face, passes the same sharpness/size gate as before, the face is steady (not moving), it differs from the last sample, and ≥ 0.35 s since the last one. It stops at the target, on Stop, or 30 s after the portal stops watching. The door's own result for each frame is computed first and cannot be changed by it.
- **Identity guard**: a sample is refused when it clearly matches *another* enrolled person at the door's own threshold ("this looks like USER5, not …"). Applies to manual and automatic capture.
- **Manual capture**: one face-detector pass instead of three (23.6 → 7.9 ms).

- **Portal**: new routes `/users/{id}/face/auto` (start / progress / stop, staff only). The Face ID panel shows progress and the server's live hint every 0.5 s. The live preview now uses the frame the face server already has (0.4 s refresh), with the camera's own snapshot only as a fallback — the camera does no extra work.
- **Removed slowness**: the old auto-capture made one camera round trip per sample, an extra status call, a 0.9 s pause, and stopped on the first rejected sample.

**Measured (TESTED OFFLINE, real models, real camera frame, temporary folder)**

- Right person (USER5): first sample saved **0.45 s** after start; a still, unchanged frame was then refused as "same as the last photo".
- Wrong name (USER5's face enrolled as a test name): refused on every frame — nothing saved.
- The real dataset was not modified (USER5 still has 23 samples).

**Firmware (ESP32-CAM)**

- Board verified before flashing: COM9 = CH340 USB-serial, boot banner "SMART LAB - ESP32-CAM STAGE 3", chip ESP32-D0WD-V3, 4 MB flash. Full flash backup taken first ( `backups/face-speed-20261014/` , local only — it contains the Wi-Fi password).
- Change in source: the manual-capture upload sends without the Windows delayed-ACK stall (same setting the door upload already uses) and logs its timing.
- **Result:** build OK and flash OK (hashes verified), but with the installed core 3.3.12 the camera fails to start ("frame buffer malloc failed") although 4 MB PSRAM is free — a camera-driver regression in that core. The working image was built with an earlier 3.3.x core (Jul 2026). The backup was written back twice and the camera boots normally (Camera OK 640×480, Wi-Fi .150, READY). The sketch now carries a warning not to build with 3.3.12.
- Automatic enrollment does **not** need new firmware: it uses the frames the camera already sends.
- Observed on the serial port (laptop unreachable): the camera reports "cannot connect", backs off, and sends no face result — so the Master gets no fresh face and cannot open on it.

## 6. Why two different latency numbers ("8×" vs "97 → 59 ms")

They measure different things. **"8×"** is the single environment function on its own (92 → 11.6 ms). **97 → 59 ms** is the whole live `/monitor` request: login check, about 15 other queries, building the tiles and sending the JSON — only one part of it got faster. Both are correct for their scope.

## 7. Acceptance matrix

| Item | Result | Evidence / reason |
|---|---|---|
| UR5e live twin (read-only) | **PASS** | LIVE-VERIFIED 14 Oct: 48 Hz, each joint checked, stale in 2.0 s |
| FANUC live twin (read-only) | **PASS** | LIVE-VERIFIED 14 Oct: twin followed pendant jogging; GET only |
| FANUC J2/J3 coupling on twin | **NOT TESTED** | needs the pose test in section 8 |
| UR5e + FANUC at the same time | **NOT TESTED** | tested only in simulation (2 sim robots) |
| Camera 1 live detection on GPU | **PASS** | LIVE-VERIFIED 14 Oct, 7 passes/s |
| Camera 1 new tracking/identity (14 Oct) | **NOT TESTED** | deployed; tested on recordings only (148 tests pass) |
| Camera 2 MC200 | **BLOCKED** | device arrives tomorrow; two-camera software tested offline |
| Cross-camera identity hand-over | **FAIL** | not implemented |
| Full-system load test | **PASS** | TESTED OFFLINE, 0 errors, see section 3 |
| Command Center refresh fix | **PASS** | TESTED OFFLINE; awaiting deploy approval |
| 3D twin smoothness in a real browser | **NOT TESTED** | headless measurement not representative |
| Door / Master ESP32 flow | **PASS** | validated earlier; unchanged |
| Robot test suite | **PASS** | 32 passed (this session) |
| Camera test suite | **PASS** | 148 passed (this session) |
| Backend test suite | **NOT TESTED** | 436 passed at the 14 Oct deploy; since then only the enrollment tests were run (12 passed). Full suite (~7 min) to run before the commit. |
| Frontend type check + build | **PASS** | after the fix (this session) |
| FANUC controller battery (DIAG-009) | **FAIL** | battery to replace — section 9 |
| **ESP32-CAM face enrollment and door recognition** | | |
| Automatic enrollment (steady, sharp, new-looking frames) | **PASS** | TESTED OFFLINE: real models + real camera frame, first sample 0.45 s; 15 face-server tests |
| Wrong person cannot be enrolled under another name | **PASS** | TESTED OFFLINE: USER5's face refused under a test name on every frame |
| Manual capture: 1 detector pass instead of 3 | **PASS** | TESTED OFFLINE: 23.6 → 7.9 ms |
| Recognition accuracy, threshold unchanged | **PASS** | TESTED OFFLINE: 97.3% right, 0 wrong person (74 samples, 5 people) |
| Portal enrollment routes + preview fallback | **PASS** | 12 backend enrollment tests on the test DB; type check + build OK |

| | | |
|---|---|---|
| Laptop analysis time per door frame | **PASS** | TESTED OFFLINE: median 48.9 ms, p95 61.8 ms |
| ESP32-CAM firmware built with core 3.3.12 | **FAIL** | camera init fails; rolled back from the backup |
| ESP32-CAM after rollback | **PASS** | LIVE (serial): Camera OK 640×480, Wi-Fi .150, READY |
| Camera with the laptop unreachable | **PASS** | LIVE (serial): "cannot connect", backs off, no face result sent |
| Live "before" timings (camera link, face server, preview) | **PASS** | LIVE: measured tonight — see section 5 |
| Live enrollment "after" timings | **BLOCKED** | needs your approval to restart the face server |
| Live failed recognition at the door (unknown face) | **NOT TESTED** | needs the network link and the door set-up |

## 8. Physical acceptance checklists

For all of these: the portal only reads. Motion is done only by the robot operator with the pendant, under the lab's normal safety rules.

### FANUC ER-4iA — J2/J3 test

1. Start the FANUC monitor; confirm the twin shows "FANUC live" with a fresh age.
2. Operator jogs **J2 only** by about +20° in JOINT mode. Check: the console reports J2 moved and **not** J3; on the twin, does the forearm keep its angle to the ground (coupling right) or rotate with the upper arm (coupling wrong)?
3. Jog **J3 only** by about +20°. Only the forearm should move on the twin.
4. Take a photo of the real arm and a screenshot of the twin in the same pose. Record the pendant joint values for both poses.
5. Repeat J1, J4, J5, J6 one at a time (direction and size).

### UR5e

1. Cable to the Smart_Lab router; robot IP 192.168.1.102; start the UR5e monitor.
2. Check rate ~48 Hz and joint directions once more after the move to the shared router.
3. Unplug the cable: the twin must show TELEMETRY_STALE within ~2 s, and recover when plugged back.

### Both robots together

1. Run both monitors; both twins fresh at the same time on /command.
2. Jog one robot: only its own twin moves. Note the portal latency and CPU with both live plus the camera.

### Camera 1 (MC210)

1. Start the camera; check GPU engines and ~7 passes/s in the metrics.
2. Walk in through the door: your name stays on you; sit next to another person: labels do not swap.
3. Sit at someone else's desk: you are not given their name; the desk guess shows only "? (desk)".

### Camera 2 (MC200 1080p)

1. Set its IP and stream on the Smart_Lab router; copy `camera2.env.example` to `camera2.env` and fill the source locally (never commit).

2. Start via `launch.bat`; check port 5101, both cameras ~7 passes/s, GPU load.
3. Note: a person moving between the two cameras is not matched yet.

## 9. DIAG-009 — FANUC controller battery

The pendant shows **DIAG-009**, which you reported as a battery warning on the controller. The exact message text was not captured — take a photo of the alarm line next time.

- Do not ignore it: if the backup battery runs flat, the controller can lose its memory, or the robot can lose its position calibration (mastering) and need re-mastering by a qualified person.
- Before replacing: make a full controller backup (image / all files) to a USB stick.
- Replace the battery the FANUC maintenance manual specifies for this controller and robot, following the manual's procedure. For FANUC robot (encoder) batteries the controller is normally kept **powered on** during replacement. Check this in the manual for your model — UNVERIFIED here.
- This is lab-staff work; the portal and this project never change controller settings.

## 10. What needs your approval

1. **Restart the face server** so automatic enrollment and the identity guard go live (door recognition is briefly unavailable during the restart).
2. **Commit** both missions: Command Center fix, load-test tools, face server, portal enrollment, firmware source fix + warning, tests, docs.
3. **Rebuild and restart** the frontend and backend containers (Command Center fix, new Face ID panel and routes).
4. **Optional:** install the earlier Arduino-ESP32 3.3.x core (Boards Manager) so the firmware fix can be flashed safely; the backup stays ready for a rollback.
5. Pushing to GitHub and the password changes remain postponed, as you asked.

**Files changed (uncommitted)**

- Performance: `frontend/src/pages/CommandCenter.tsx`, `tools/perf/load_simlab.py`, `tools/perf/ws_clients.mjs`, `docs/perf/LOAD_TEST.md`.
- Face: `firmware/face_server/face_server.py`, `firmware/face_server/test_auto_enroll.py`, `firmware/face_server/bench_face_pipeline.py`, `backend/app/api/routes/enrollment.py`, `backend/tests/test_enrollment.py`, `frontend/src/components/FaceEnrollment.tsx`, `frontend/src/lib/api.ts`, `firmware/ESP32CAM_Vision_FAST/ESP32CAM_Vision_FAST.ino` (source only, not on the board).
- Docs: `docs/PROGRESS_CHECKPOINT.md`, this report.

## 11. Next

- With your approval: restart the face server, then measure live enrollment with you in front of the camera (adds real samples to your profile), door decision time, an unknown face, and the camera unplugged.
- Physical test: move the ESP32-CAM 2–3 m from the Smart_Lab router and re-measure the 780 ms frame trip.
- Remaining door bottleneck: the camera's capture + Wi-Fi upload (~0.5 s per frame). With the much stronger signal now seen at the router (−19 to −27 dBm vs −65 to −75 dBm at the door before), two frames in flight may help — only to be tried and measured at the door's real position.

- At the lab: FANUC J2/J3 test, UR5e on the shared router, both robots together, camera 1 identity check, MC200 setup.